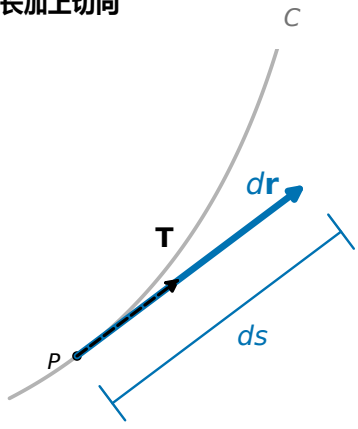
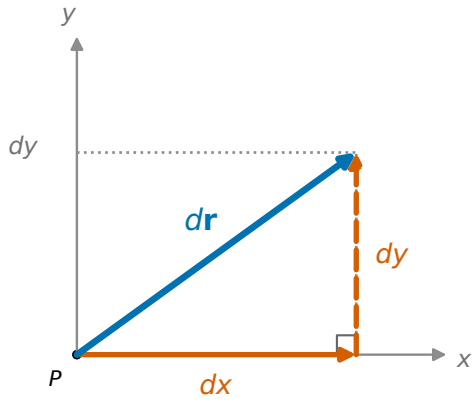
**A 弧长加上切向**




$$d\mathbf{r} = \mathbf{T}\,ds, \quad \|\mathbf{T}\| = 1$$

**B 坐标投影与分量合成**




$$d\mathbf{r} = dx\,\mathbf{e}_x + dy\,\mathbf{e}_y$$